

Module 2

Relational Model and E-R Modeling

Topic

Entity Relationship Model : Types of Attributes, Relationship, Structural Constraints

(Abraham Silberschatz)

Entity-Relationship (E-R) data model

- The **entity-relationship (E-R) data model** was developed to **facilitate database** design by allowing specification of an *enterprise schema that represents the overall logical structure of a database.*
- The E-R model is very useful in **mapping the meanings and interactions of real-world enterprises** onto a conceptual schema



Modeling

- A *database* can be modeled as:
 - a collection of entities,
 - relationship among entities.
- An **entity** is an object that exists and is distinguishable from other objects.
 - Example: specific person, company, event, plant
- Entities have **attributes**
 - Example: people have *names* and *addresses*
- An **entity set** is a set of entities of the same type that share the same properties.
 - Example: set of all persons, companies, trees, holidays



Entity Sets *instructor* and *student*

instructor_ID instructor_name

76766	Crick
45565	Katz
10101	Srinivasan
98345	Kim
76543	Singh
22222	Einstein

instructor

student-ID student_name

98988	Tanaka
12345	Shankar
00128	Zhang
76543	Brown
76653	Aoi
23121	Chavez
44553	Peltier

student

Relationship

Relationship Sets

- A **relationship** is an association among several entities

Example:

44553 (Peltier)	<u><i>advisor</i></u>	22222 (<u>Einstein</u>)
<i>student</i> entity	relationship	<i>instructor</i> entity

- A **relationship set** is a mathematical relation among $n \geq 2$ entities, each taken from entity sets

$$\{(e_1, e_2, \dots, e_n) \mid e_1 \in E_1, e_2 \in E_2, \dots, e_n \in E_n\}$$

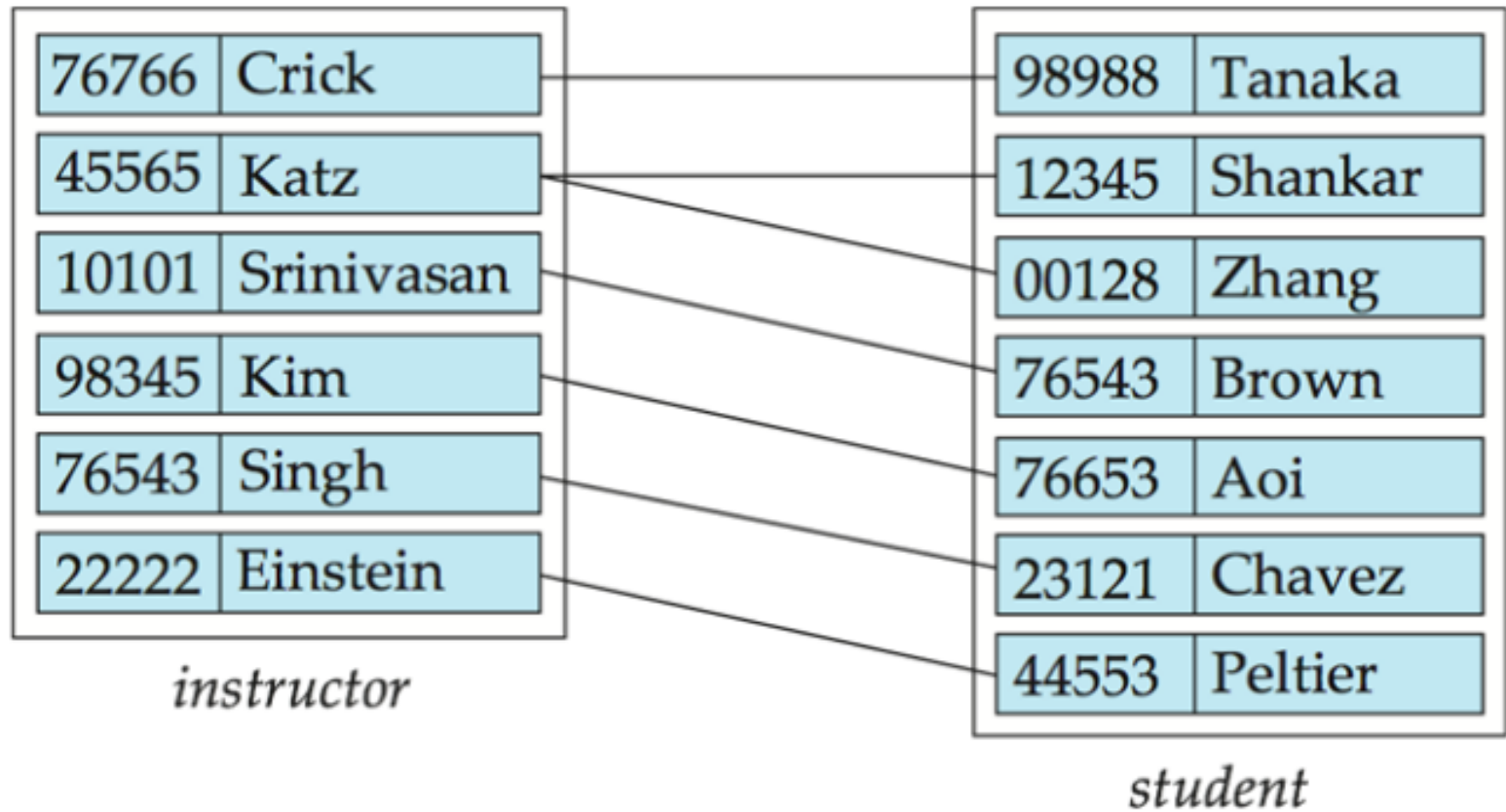
where $(e_1, e_2, \dots, e_n)$ is a relationship

- Example:

$(44553, 22222) \in \text{advisor}$



Relationship Set *advisor*



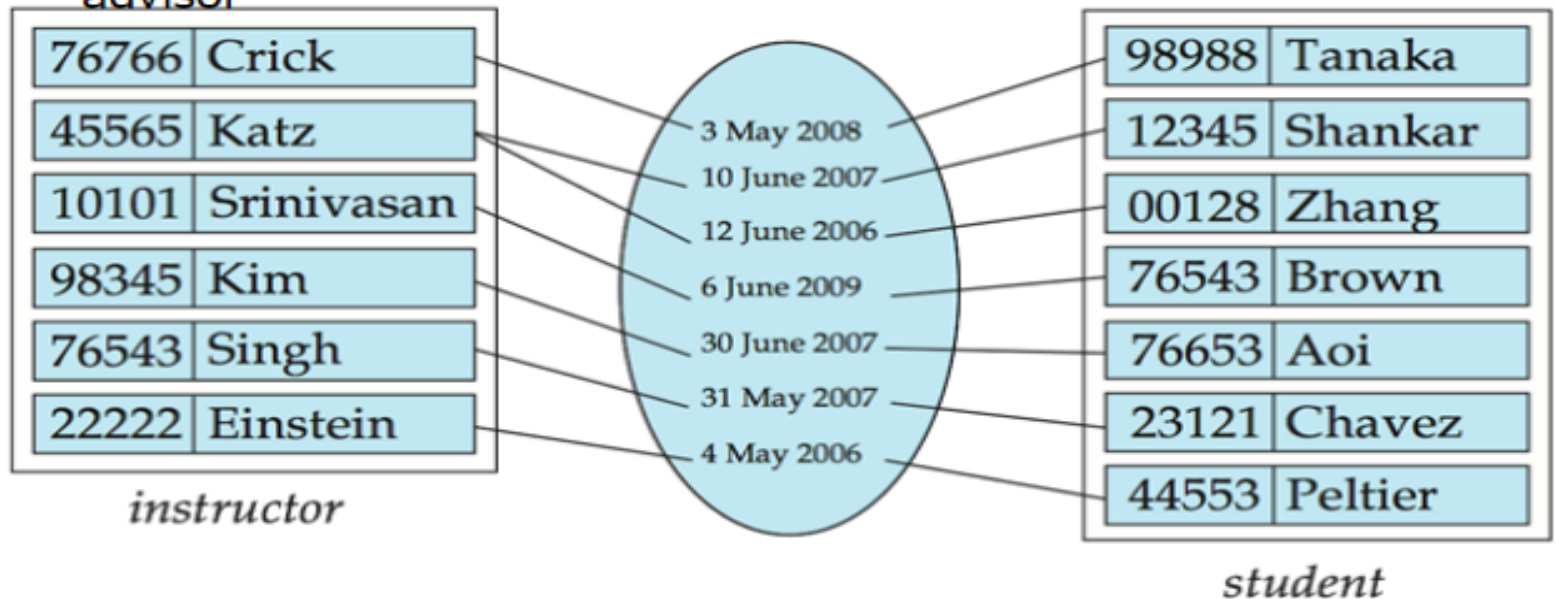
Relationship

- The association between entity sets is referred to as **participation**; that is, the **entity sets $E1, E2, \dots, En$ participate in relationship set R** .
- A **relationship instance** in an E-R schema represents an association between the named entities in the real-world enterprise that is being modeled.
- As an illustration, the individual *instructor entity Katz, who has instructor ID 45565, and the student entity Shankar, who has student ID 12345, participate in a relationship instance of advisor*.
- *This relationship instance represents that in the university, the instructor Katz is advising student Shankar.*

Relationship Sets (Cont.)

An **attribute** can also be property of a relationship set.

- For instance, the *advisor* relationship set between entity sets *instructor* and *student* may have the attribute *date* which tracks when the student started being associated with the advisor



A relationship may also have attributes called **descriptive attributes**.



Degree of a Relationship Set

- **binary relationship**
 - involve two entity sets (or degree two).
 - most relationship sets in a database system are binary.
- Relationships between more than two entity sets are rare. (More on this later.)
 - ▶ Example: *students* work on research *projects* under the guidance of an *instructor*.
 - ▶ relationship *proj_guide* is a ternary relationship between *instructor*, *student*, and *project*

Types of Attributes

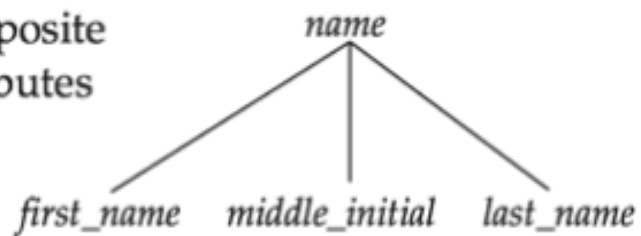
Attributes

- An entity is represented by a set of attributes, that is descriptive properties possessed by all members of an entity set.
 - Example:
instructor = (ID, name, street, city, salary)
course = (course_id, title, credits)
- **Domain** – the set of permitted values for each attribute
- Attribute types:
 - **Simple** and **composite** attributes.
 - **Single-valued** and **multivalued** attributes
 - Example: multivalued attribute: *phone_numbers*
 - **Derived** attributes
 - Can be computed from other attributes
 - Example: age, given date_of_birth

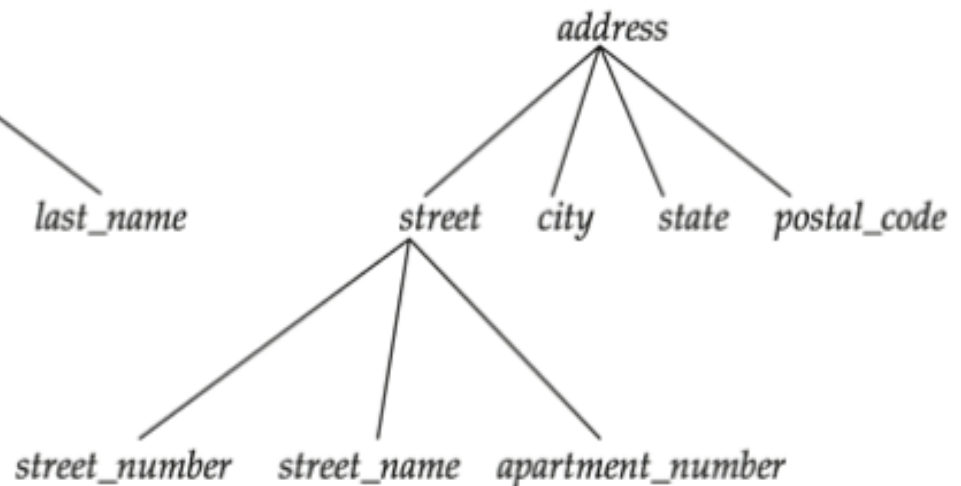


Composite Attributes

composite
attributes



component
attributes



Structural Constraints

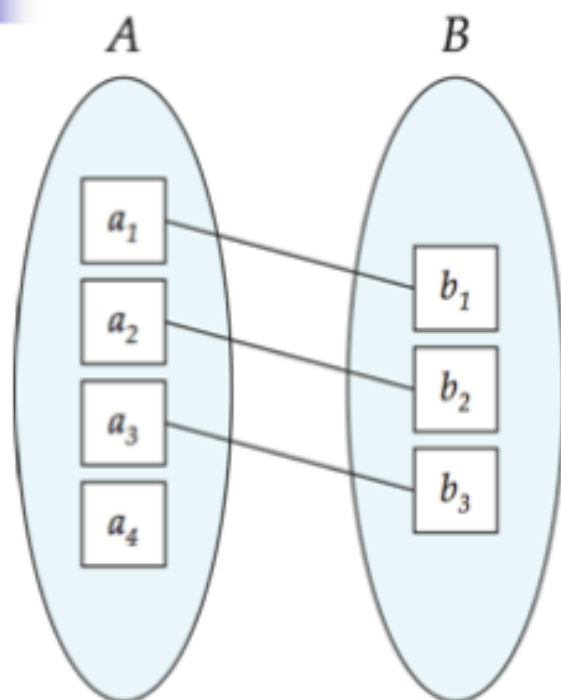


Mapping Cardinality Constraints

- Express the number of entities to which another entity can be associated via a relationship set.
- Most useful in describing binary relationship sets.
- For a binary relationship set the mapping cardinality must be one of the following types:
 - One to one
 - One to many
 - Many to one
 - Many to many

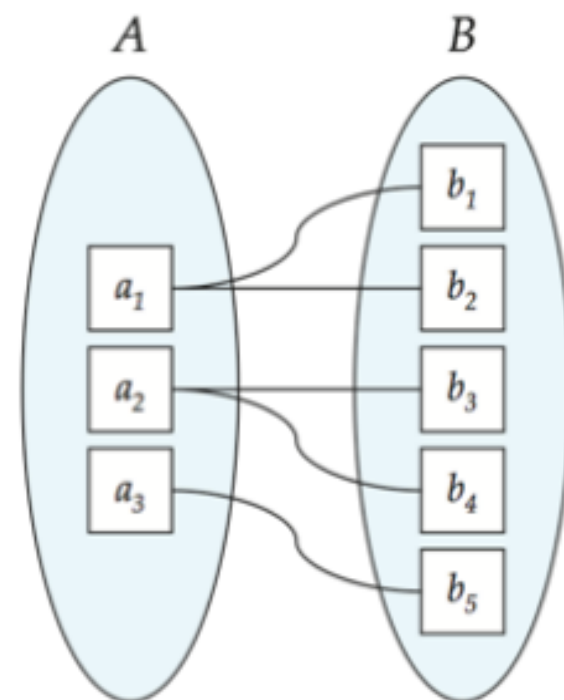


Mapping Cardinalities



(a)

One to one

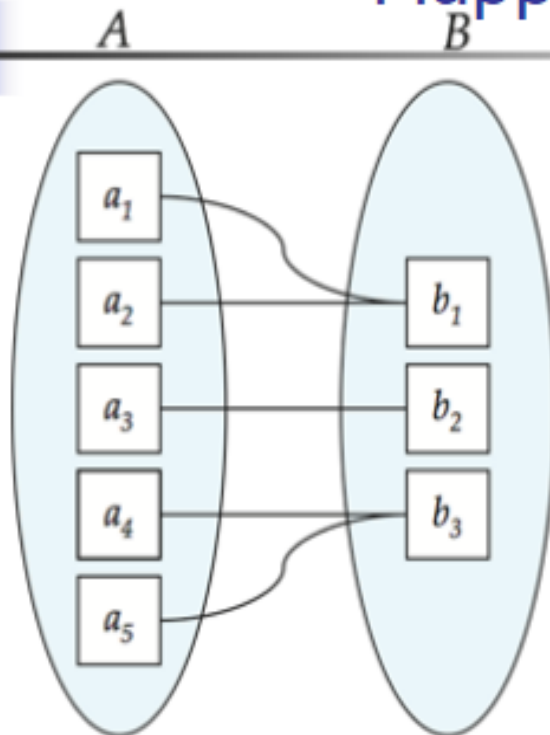


(b)

One to many

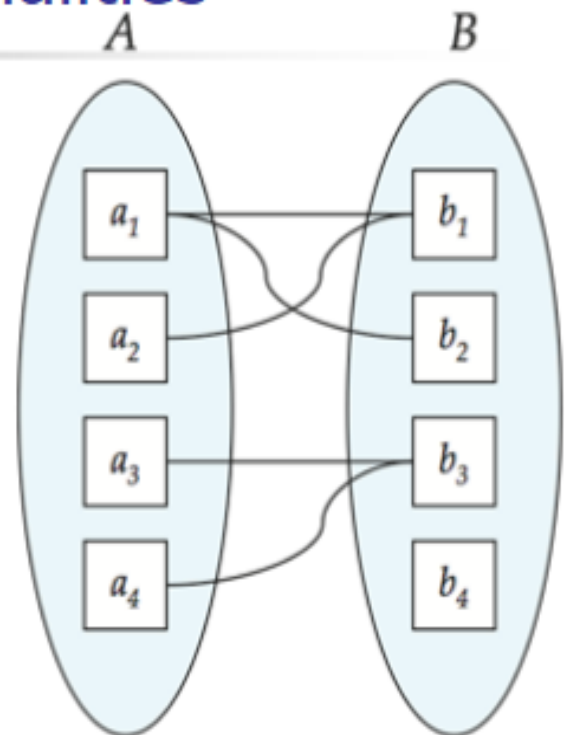
Note: Some elements in *A* and *B* may not be mapped to any elements in the other set

Mapping Cardinalities



(a)

Many to one



(b)

Many to many

Note: Some elements in A and B may not be mapped to any elements in the other set

Participation Constraints

- The participation of an entity set E in a relationship set R is said to be **total** if **every entity** in E participates in at least one relationship in R .
- If **only some entities** in E participate in relationships in R , the participation of entity set E in relationship R is said to be **partial**.



Keys

- A **super key** of an entity set is a set of one or more attributes whose values uniquely determine each entity.
- A **candidate key** of an entity set is a minimal super key
 - *ID* is candidate key of *instructor*
 - *course_id* is candidate key of *course*
- Although several candidate keys may exist, one of the candidate keys is selected to be the **primary key**.

- **Super Key:**

Super Key is an attribute (or set of attributes) that is used to uniquely identifies all attributes in a relation.

- All super keys can't be candidate keys but its reverse is true.
- In a relation, number of super keys are more than number of candidate keys.
- **Example:** We have a given relation $R(A, B, C, D, E, F)$ and we shall check for being super keys by following given dependencies:

Functional dependencies	Super key
AB- $\rightarrow$ CDEF, CD- $\rightarrow$ ABEF	YES
CB- $\rightarrow$ DF , D- $\rightarrow$ BC	NO

- **Candidate Key:** A candidate key is a set of attributes (or attributes) that uniquely identify the tuples in relation to or table.
- The candidate key's attributes can contain a NULL value which opposes to the primary key.
- **Example:**
- Student{ID, First_name, Last_name, Age, Sex, Phone_no}

- **Primary Key:**
- Candidate key is a set of attributes (or attribute) which uniquely identify the tuples in a relation or table.
- There can be more than one candidate key in relation out of which **one can be chosen as the primary key.**
- **Example:** Student{Stud_No, Stud_name, Stud_phone, Stud_state, Stud_country, Stud_age}
- Here we can see the two candidate keys **Stud_No** and **Stud_phone.**
- STUD_No **can be chosen as the primary key** (only one out of many candidate keys).



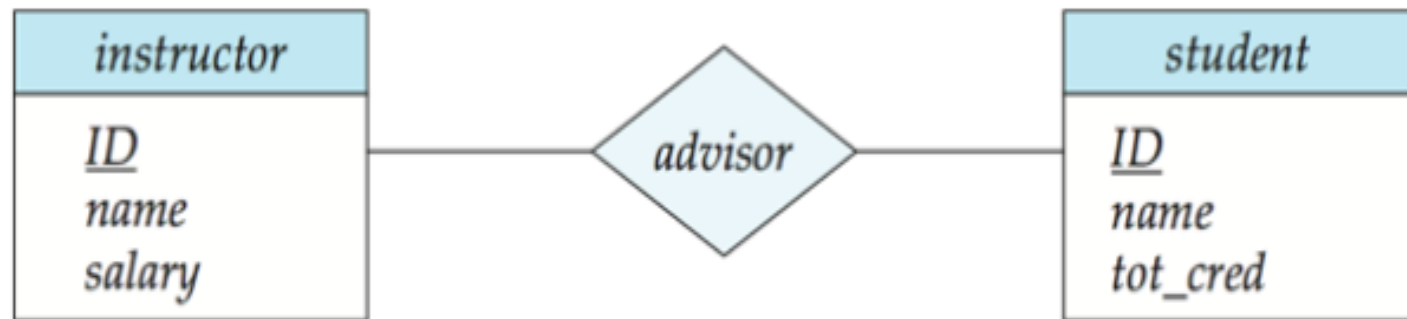
Redundant Attributes

- Suppose we have entity sets
 - *instructor*, with attributes including *dept_name*
 - *department*and a relationship
 - *inst_dept* relating *instructor* and *department*
- Attribute *dept_name* in entity *instructor* is redundant since there is an explicit relationship *inst_dept* which relates instructors to departments
 - The attribute replicates information present in the relationship, and should be removed from *instructor*
 - BUT: when converting back to tables, in some cases the attribute gets reintroduced, as we will see.

Removing Redundant Attributes in Entity Sets

- The entity set *instructor* includes the attributes *ID*, *name*, *dept name*, and *salary*, with *ID* forming the primary key.
- The entity set *department* includes the attributes *dept name*, *building*, and *budget*, with *dept name* forming the primary key.
- We model the fact that each instructor has an associated department using a relationship set *inst_dept* relating *instructor* and *department*.
- The attribute *dept name* appears in both entity sets. Since it is the primary key for the entity set *department*, it is *redundant in the entity set instructor and needs to be removed*.

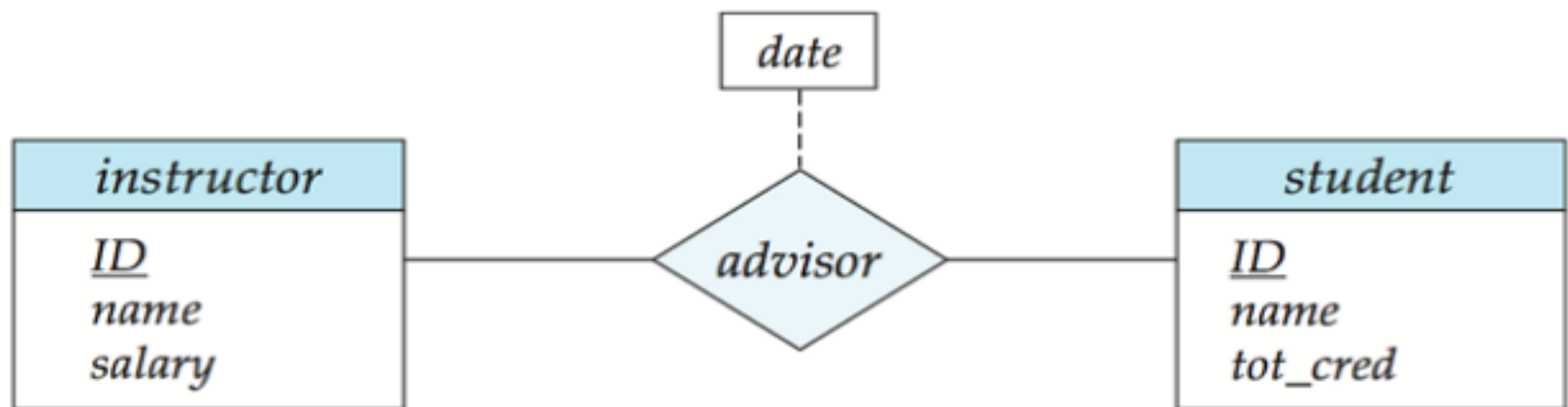
E-R Diagrams



- n Rectangles represent entity sets.
- n Diamonds represent relationship sets.
- n Attributes listed inside entity rectangle
- n Underline indicates primary key attributes



Relationship Sets with Attributes



E-R Diagrams

- An E-R diagram can express the overall logical structure of a database graphically. **E-R diagrams are simple and clear—qualities** that may well account in large part for the widespread use of the E-R model.
- **Rectangles divided into two parts represent entity sets.** The first part, which in this textbook is shaded blue, contains the name of the entity set. The second part contains the names of all the attributes of the entity set.
- **Diamonds** represent relationship sets.
- **Undivided rectangles** represent the attributes of a relationship set.
- **Attributes** that are part of the primary key are underlined.
- **Lines link** entity sets to relationship sets.
- **Dashed lines link** attributes of a relationship set to the relationship set.
- **Double lines** indicate total participation of an entity in a relationship set.
- **Double diamonds** represent identifying relationship sets linked to weak entity sets

Entity With Composite, Multivalued, and Derived Attributes



instructor

ID

name

first_name

middle_initial

last_name

address

street

street_number

street_name

apt_number

city

state

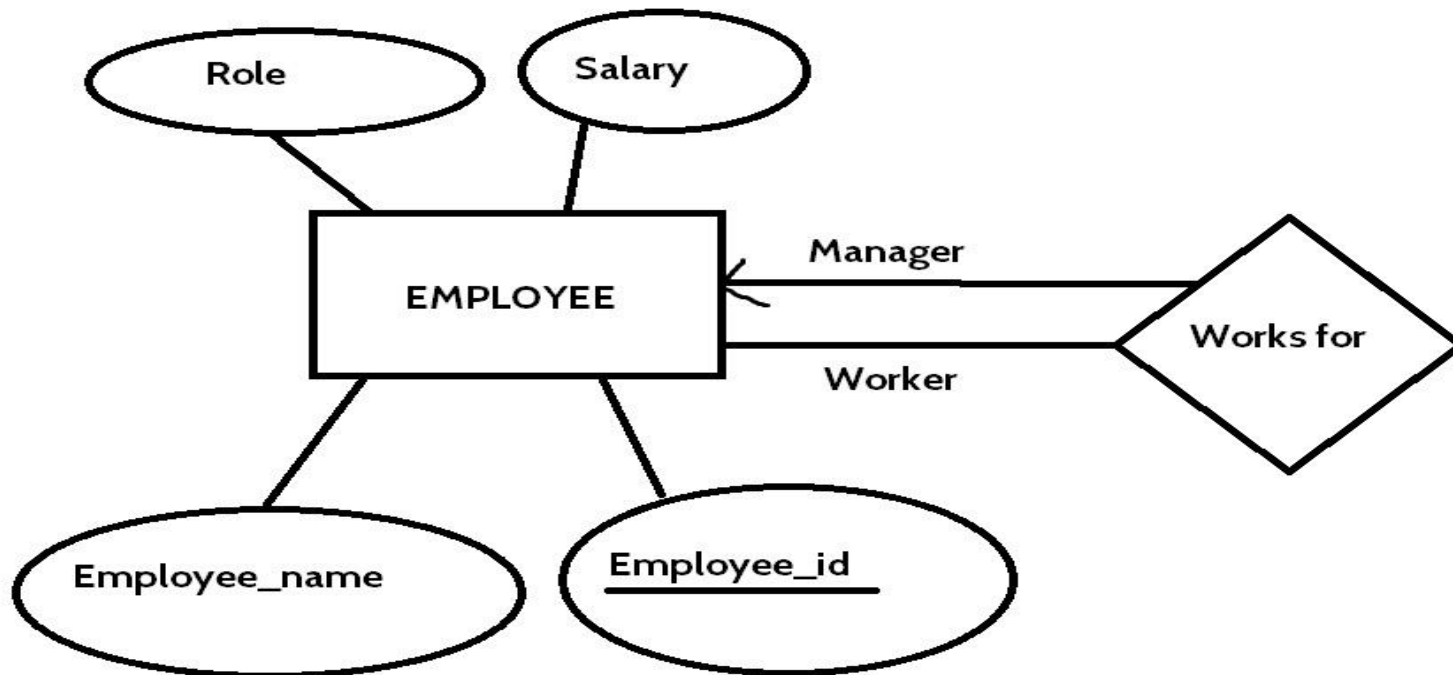
zip

{ *phone_number* }

date_of_birth

age ()

E-R diagram with role indicators



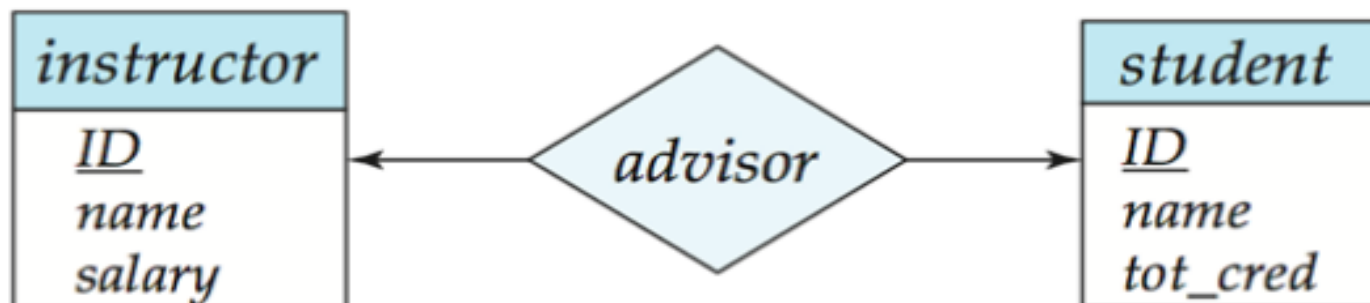


Cardinality Constraints

- We express cardinality constraints by drawing either a directed line ($\rightarrow$), signifying "one," or an undirected line ($—$), signifying "many," between the relationship set and the entity set.
- One-to-one relationship:
 - A student is associated with at most one *instructor* via the relationship *advisor*
 - A *student* is associated with at most one *department* via *stud_dept*

One-to-One Relationship

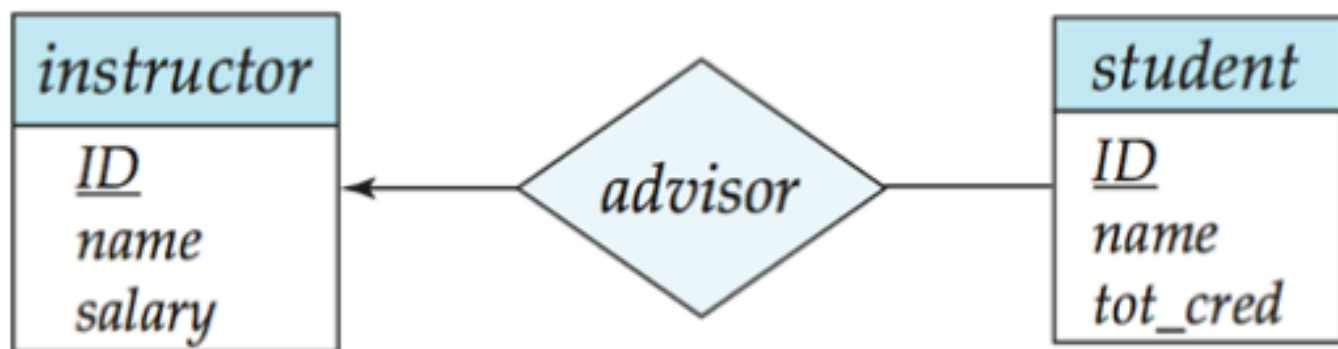
- one-to-one relationship between an *instructor* and a *student*
 - an instructor is associated with at most one student via *advisor*
 - and a student is associated with at most one instructor via *advisor*





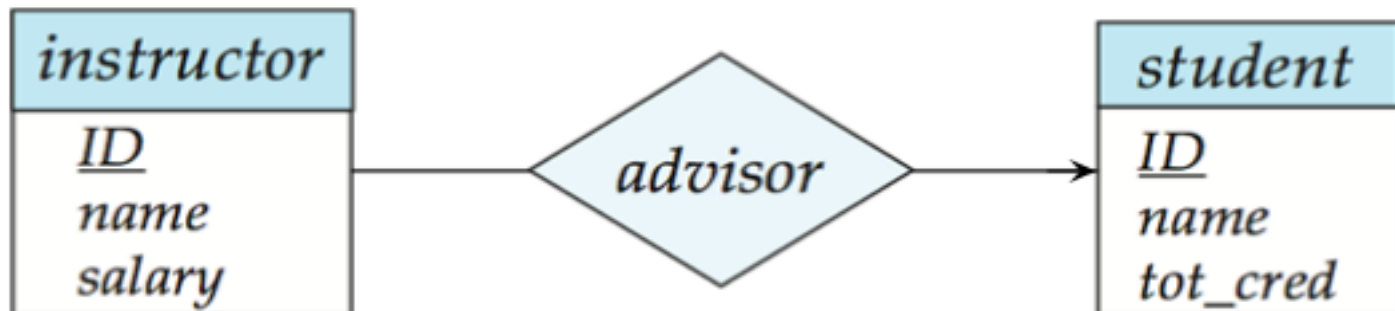
One-to-Many Relationship

- one-to-many relationship between an *instructor* and a *student*
 - an instructor is associated with several (including 0) students via *advisor*
 - a student is associated with at most one instructor via advisor,



Many-to-One Relationships

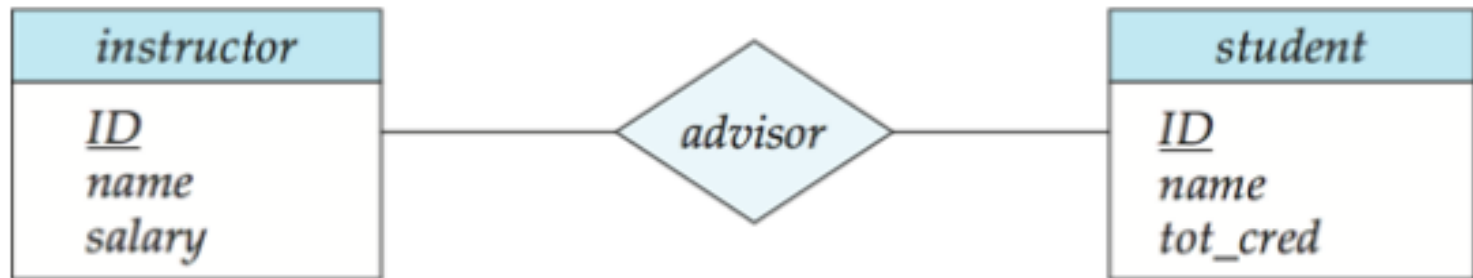
- In a many-to-one relationship between an *instructor* and a *student*,
 - an instructor is associated with at most one student via *advisor*,
 - and a student is associated with several (including 0) instructors via *advisor*





Many-to-Many Relationship

- An instructor is associated with several (possibly 0) students via *advisor*
- A student is associated with several (possibly 0) instructors via *advisor*



Cardinality limits on relationship sets.

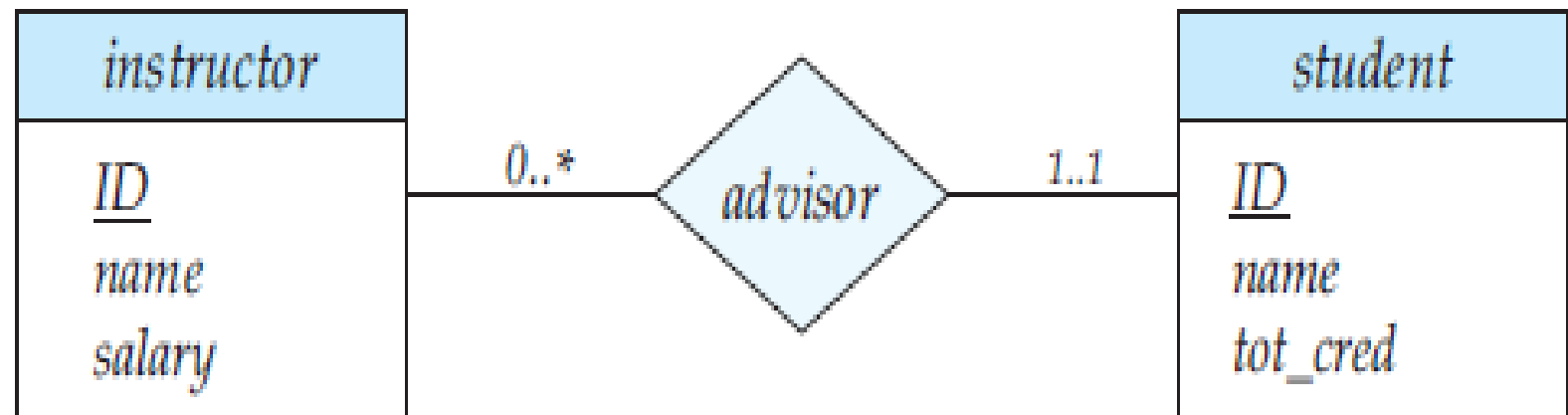
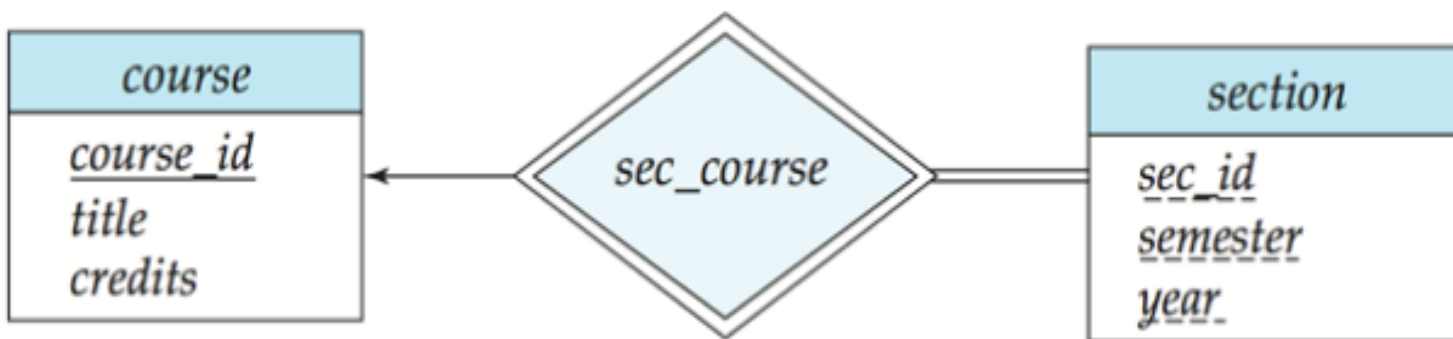


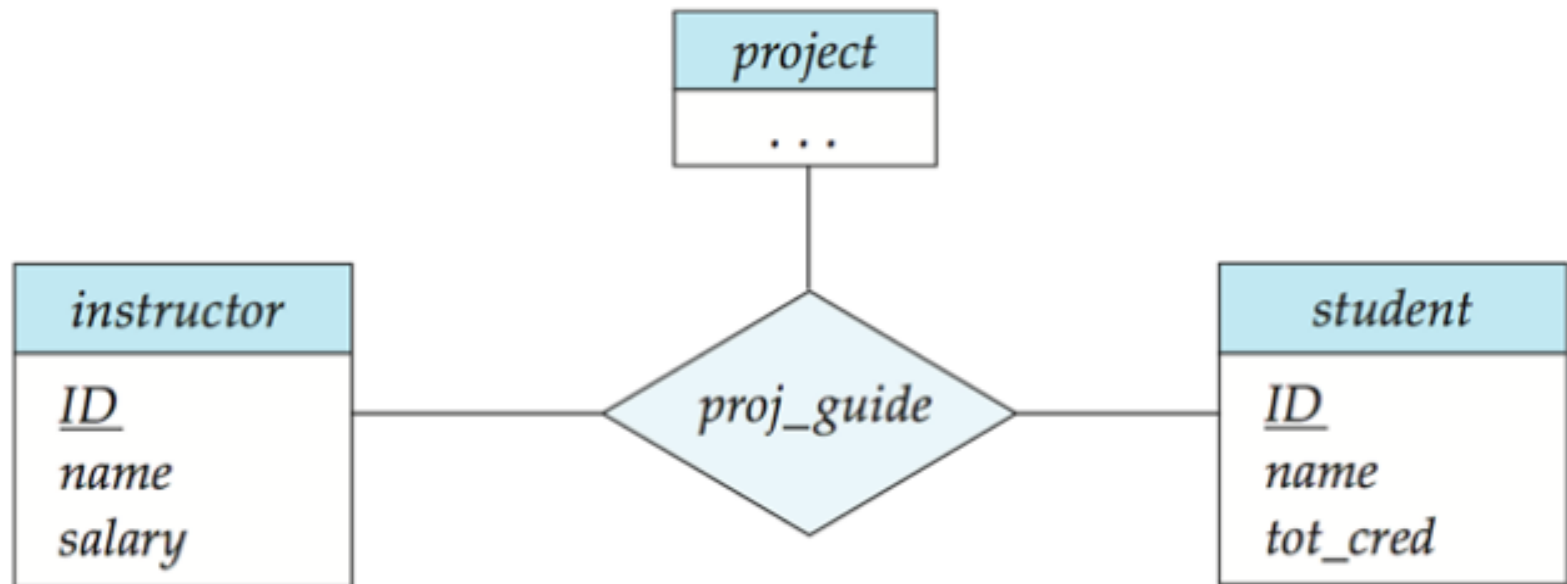
Figure 7.10 Cardinality limits on relationship sets.

Participation of an Entity Set in a Relationship Set

- n Total participation (indicated by double line): every entity in the entity set participates in at least one relationship in the relationship set
 - | E.g., participation of *section* in *sec_course* is total
 - ▶ every *section* must have an associated course
- n Partial participation: some entities may not participate in any relationship in the relationship set
 - | Example: participation of *instructor* in *advisor* is partial



E-R Diagram with a Ternary Relationship





Weak Entity Sets

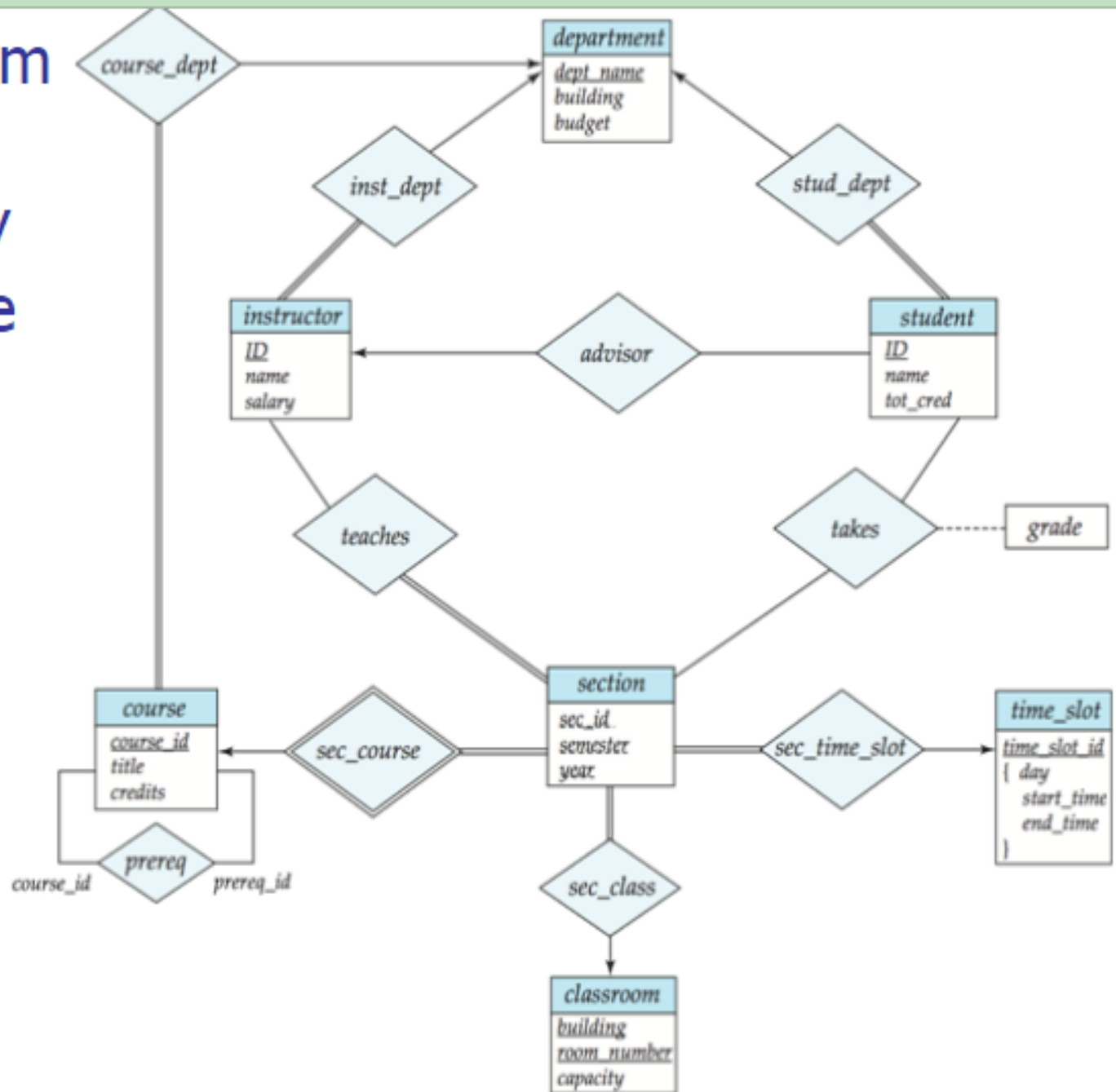
- An entity set that does not have a primary key is referred to as a **weak entity set** .
- The existence of a weak entity set depends on the existence of a **identifying entity set**
 - It must relate to the identifying entity set via a total, one-to-many relationship set from the identifying to the weak entity set
 - **Identifying relationship** depicted using a double diamond
- The **discriminator** (*or partial key*) of a weak entity set is the set of attributes that distinguishes among all the entities of a weak entity set.
- The primary key of a weak entity set is formed by the primary key of the strong entity set on which the weak entity set is existence dependent, **plus** the weak entity set's discriminator.

Weak Entity Sets (Cont.)

- We underline the discriminator of a weak entity set with a dashed line.
- We put the identifying relationship of a weak entity in a double diamond.
- Primary key for *section* – (*course_id*, *sec_id*, *semester*, *year*)



E-R Diagram for a University Enterprise





In-Class Exercises

7.15 Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted.

7.17 Design an E-R diagram to keep track of games played, scores of each game, players in each game, and player statistics for each game for **all teams in a league**. Summary statistics should be modeled as derived attributes. (cf. HW 7.3)